Tezpur University

Spring Semester End examination-2023

**Course Code: CO218** **Course Title: Data Communication**

*Full marks=60* *Time=3 hours*

*(The figures in the right-hand margin indicate full marks for the questions)*

1. Answer the following to the point? (Answer any **five**) **[2x5=10]**
   a. Write the difference between static and dynamic routing.
   b. Write the difference between MACA and MACAW
   c. Why is the statistical TDM important over TDM link?
   d. Name two encapsulated forms of PPP for DLC
   e. Write the difference between IEEE 802.4 and IEEE 802.5
   f. Name the different types of configurations supported in IEEE 802.1Q
   g. Write the difference between bridge and switch

2. In what situation collision is still possible in CSMA/CD? Why is CSMA/CD not possible to be used in wireless LAN? **[2+1=3]**

3. State the principle of bidding and conflict avoidance rule of binary count down protocol. **[2+2=4]**

4. What is the key principle of code division multiplexing? List the features of the key principle. **[1+2=3]**

5. What is the working principle of *truncated binary exponential backoff (BEF)* algorithm used in CSMA? **[3]**

6. How does HDLC handle the situations of i) busy condition and ii) time-out recovery? **[2+2=4]**

7. An MFSK scheme sends data 2 bits at a time at a bit rate of 2Mbps. If the carrier frequency is 10MHz, find the number of different frequencies required and the bandwidth. **[3]**

8. What are the different data rate matching techniques used in synchronous TDM? Discuss about them briefly? **[4]**

9. A time division multiplexer multiplexes four 100 bps channels. If the MUX multiplexes 1 byte per channel, compute the frame size, rate, duration and the bit rate of the link. **[4]**

10. Consider a stream of binary 1s is required to be transmitted with a bit duration of 1 μs each using NRZ-I and Manchester encoding. If 2 number of bits per signal element needed to be encoded, calculate the modulation rates for both the schemes. **[4]**

11. Discuss about different types of operational modes implemented by IEEE 802.2 standard. **[4]**

Or

12. Discuss about the working principle of distance vector routing and link state routing algorithms. **[4]**

13. Consider four digital data channels each transmitting at 1 Mbps speed. Design an appropriate configuration for FDM, using a satellite channel of bandwidth 1 MHz. **[4]**

14. Consider for a given 10kbps data rate line, the receiver is faster by 6% per bit time. The transmitter employs 8-bit character asynchronous transmission scheme and requires 100 μsecs to transmit a character. Assume the transmitter starts transmitting the start bit at time 50th μsec. (i) Calculate the transmission time for each of the bits of the character. (ii) Calculate the sampling rate of the received character. **[4]**

15. Answer the following questions appropriately. (Answer any **three**) **[2x3=6]**
   a. Draw the waveform to encode the data string "01000101011000000011" using B8ZS encoding scheme
   b. State the role of P-bit and F-bit in HDLC protocol
   c. State the hidden terminal and exposed terminal problems
   d. Name control signaling virtual channel connection (VCC) implemented by ATM networks